

Divisibilities

Problem 1. The natural numbers a and b are such that $\frac{a+1}{b} + \frac{b+1}{a}$ is an integer. Show that the greatest common divisor of a and b is not greater than $\sqrt{a+b}$.

Problem 2. Does there exist an infinite set of integers S such that for every $a, b \in S$ we have $a^2 + b^2 - ab \mid (ab)^2$?

Problem 3. Determine all positive integers n for which there exists a positive integer d with the property that n is divisible by d and $n^2 + d^2$ is divisible by $d^2n + 1$.

Problem 4. Find all positive integers n satisfying the following condition: For any positive integers a and b , if $n \mid a^2b + 1$, then $n \mid a^2 + b$.

Problem 5. Find all odd positive integers $n > 1$ such that if a and b are relatively prime divisors of n , then $a + b - 1$ divides n .

Problem 6. Find all positive integers n such that for any positive integer a relatively prime to n , $2n^2 \mid a^n - 1$.

Problem 7. Find all positive integers $n, k, a_1, a_2, \dots, a_k$ such that $n^{k+1} + 1$ is divisible by $(na_1 + 1)(na_2 + 1) \cdots (na_k + 1)$.

Problem 8. Prove that for any odd prime number p , the number of positive integer n satisfying $p \mid n! + 1$ is less than or equal to $cp^{\frac{2}{3}}$, where c is a constant independent of p .